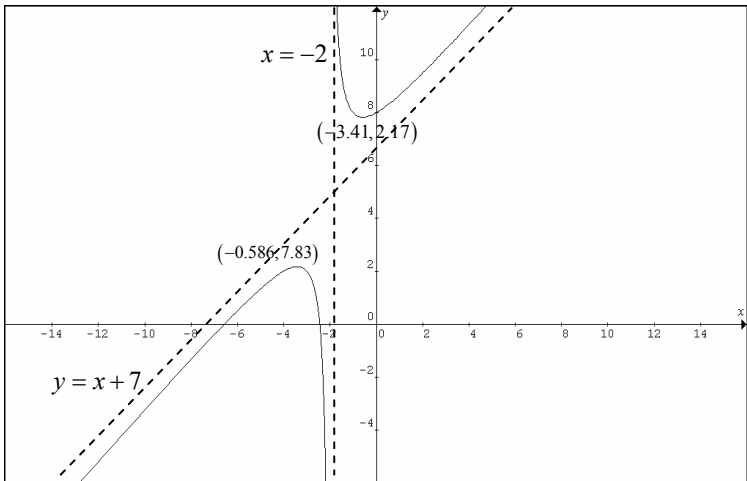
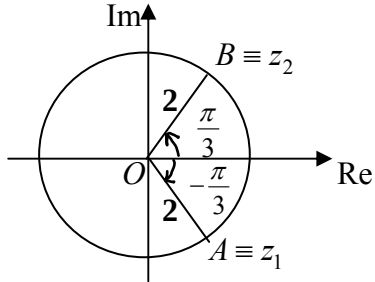
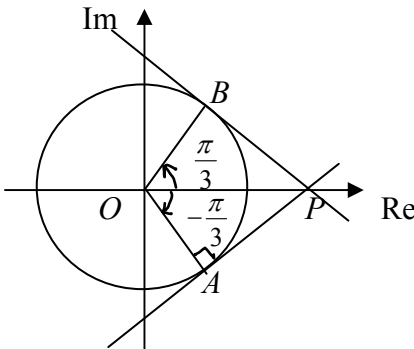
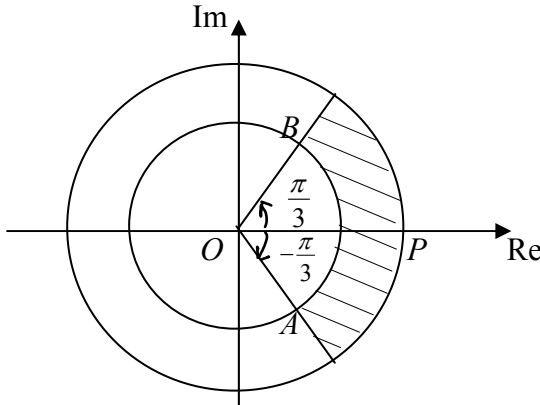


2009 Innova Junior College Preliminary Examination: H2 Maths

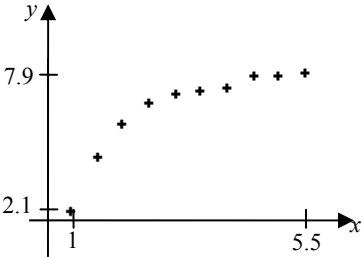
Qn	Answers
1(i)	Asymptotes: $y = x + 7$ and $x = -2$
1(ii)	$5 - 2\sqrt{2} < k < 5 + 2\sqrt{2}$
1(iii)	
2(a)	shortest distance from the origin O to the plane $\Pi = \frac{ \mathbf{r} \cdot \mathbf{n} }{ \mathbf{n} } = \frac{-14}{\left(\frac{7}{6}\right)} = 12$ units
2(b)	Let F be the foot of perpendicular from O to AB. $\overrightarrow{OF} = \mathbf{a} + \lambda(\mathbf{b} - \mathbf{a})$ Since OF is perpendicular to the line AB, $\overrightarrow{OF} \cdot (\mathbf{b} - \mathbf{a}) = 0$ $[\mathbf{a} + \lambda(\mathbf{b} - \mathbf{a})] \cdot (\mathbf{b} - \mathbf{a}) = 0$ $\mathbf{a} \cdot (\mathbf{b} - \mathbf{a}) + \lambda(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{b} - \mathbf{a}) = 0$ $\mathbf{a} \cdot \mathbf{b} - \mathbf{a} ^2 + \lambda(\mathbf{b} ^2 - 2\mathbf{a} \cdot \mathbf{b} + \mathbf{a} ^2) = 0$ $- \mathbf{a} ^2 + \lambda(\mathbf{b} ^2 + \mathbf{a} ^2) = 0$ $\lambda = \frac{ \mathbf{a} ^2}{ \mathbf{a} ^2 + \mathbf{b} ^2}$ $\overrightarrow{OF} = \mathbf{a} + \lambda(\mathbf{b} - \mathbf{a})$ $= \mathbf{a} + \frac{ \mathbf{a} ^2}{ \mathbf{a} ^2 + \mathbf{b} ^2}(\mathbf{b} - \mathbf{a})$

Qn	Answers
3(i)	$u = e^x \Rightarrow \frac{du}{dx} = e^x \Rightarrow \frac{du}{dx} = u$ $\int_0^{\ln 2} \frac{e^x}{e^x + 3e^{-x}} dx = \int_1^2 \frac{u}{u^2 + 3} du = \frac{1}{2} \left[\ln(u^2 + 3) \right]_1^2 = \frac{1}{2} \ln\left(\frac{7}{4}\right)$
3(ii)	$\text{Area} = \int_0^{\ln 2} \frac{7e^x}{e^x + 3e^{-x}} dx - \frac{1}{2}(\ln 2)(4) = \frac{7}{2} \ln\left(\frac{7}{4}\right) - 2 \ln 2$
3(iii)	$\text{Vol}_{(x)} = \pi \int_0^{\ln 2} \left(\frac{7e^x}{e^x + 3e^{-x}} \right)^2 dx - \frac{1}{3} \pi (4)^2 (\ln 2) = 6.72 \text{ unit}^3 \quad (3\text{s.f})$
3(iv)	$y = \frac{7e^{\frac{1}{3}x}}{e^{\frac{1}{3}x} + 3e^{-\frac{1}{3}x}} - 5$
4(i)	$\sum_{j=0}^3 \left(-\frac{z}{2} \right)^j = 1 - \frac{z}{2} + \left(-\frac{z}{2} \right)^2 + \left(-\frac{z}{2} \right)^3 = \frac{1 \left(1 - \left(-\frac{z}{2} \right)^4 \right)}{1 + \frac{z}{2}} = \frac{\left(\frac{16 - z^4}{16} \right)}{\frac{2 + z}{2}}$ $= \frac{16 - z^4}{16 + 8z}$
4(ii)	$\sum_{j=0}^3 \left(-\frac{z}{2} \right)^j - 1 = 0 \Rightarrow \frac{16 - z^4}{16 + 8z} = 1 \Rightarrow 16 - z^4 = 16 + 8z$ $\Rightarrow z(z^3 + 8) = 0$ $z = 0 \quad \text{or} \quad z^3 + 8 = 0$ $(\text{reject since } z \neq 0) \quad z^3 = -8$ $z^3 = 8e^{i(\pi + 2k\pi)}$ $z = 2e^{\frac{1}{3}(\pi + 2k\pi)i} \quad \text{where } k = -1, 0, 1.$ $\text{Since } z \neq -2, \quad z_1 = 2e^{-\frac{\pi}{3}i}, \quad z_2 = 2e^{\frac{\pi}{3}i}.$
	Alternative: $\sum_{j=0}^3 \left(-\frac{z}{2} \right)^j - 1 = 0 \Rightarrow \frac{16 - z^4}{16 + 8z} = 1 \Rightarrow 16 - z^4 = 16 + 8z$ $z(z^3 + 8) = 0$ $z = 0 \quad \text{or} \quad z^3 + 8 = 0$ $(\text{reject since } z \neq 0) \quad (z + 2)(z^2 - 2z + 4) = 0$ $\text{Since } z \neq -2, \quad z = \frac{2 \pm \sqrt{4 - 4(1)(4)}}{2} = 1 \pm i\sqrt{3}$ $\therefore z_1 = 2e^{-\frac{\pi}{3}i}, \quad z_2 = 2e^{\frac{\pi}{3}i}$

Qn	Answers
4(iii)	 $w = 2 \Rightarrow \therefore k = 2$
4(iv)	 $\cos\left(\frac{\pi}{3}\right) = \frac{2}{s} \Rightarrow s = 4$
4(v)	$r \leq w \leq s \Rightarrow 2 \leq w \leq 4$ $\arg(z_1) \leq \arg(w) \leq \arg(z_2) \Rightarrow -\frac{\pi}{3} \leq \arg(w) \leq \frac{\pi}{3}$  Area of shaded region = $\frac{1}{3} \left[\pi (4)^2 - \pi (2)^2 \right] = 4\pi \text{ units}^2$
5	E _ _ R (M, D, I, T, EE, R, AA, NN) Case 1: The letters in the centre are different No of ways = $8 \times 7 = 56$ Case 2: Both letters are the same No of ways = 3 Total no of ways = $56 + 3 = 59$

Qn	Answers
6	Let X be the random variable “number of light bulbs to be replaced in a day” $X \sim \text{Po}(\mu)$ $P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-\mu} \frac{\mu^0}{0!}$ $1 - e^{-\mu} \frac{\mu^0}{0!} = 0.3$ $e^{-\mu} = 0.7$ $\mu = -\ln 0.7$
7(i)	No. Because not every individual has an equal chance of being selected. Only the “even numbered” students will have a chance of being selected.
7(ii)	Find sampling interval $k = \frac{2000}{40} = 50$. Get a list of all students arranged according to the school register. Select the first student at random from the first 50 students. Select every 50th student thereafter till a sample of 40 is obtained.
8(i)	$P(Y) = \left(\frac{10}{25}\right)\left(\frac{8}{12}\right) + \left(\frac{15}{25}\right)\left(\frac{3}{10}\right) = \left(\frac{67}{150}\right)$
8(ii)	$P(X \cup Y) = P(X) + P(Y) - P(X \cap Y) = \frac{10}{25} + \frac{67}{150} - \left(\frac{10}{25}\right)\left(\frac{8}{12}\right) = \frac{29}{50}$
8(iii)	$P(X Y) = \frac{P(X \cap Y)}{P(Y)} = \frac{\frac{4}{15}}{\frac{67}{150}} = \frac{40}{67}$ $P(X) = \frac{2}{5} \neq P(X Y)$ Hence events X and Y are not independent.
9(i)	Let D be the random variable “mass of a durian, in kg.” $D \sim N(1.5, 0.2^2)$ Let M be the random variable “mass of a mango, in kg.” $M \sim N(0.4, 0.05^2)$ $M_1 + M_2 + M_3 + M_4 + M_5 \sim N(2, 0.0125) \quad 2(M_1 + M_2 + M_3 + M_4 + M_5) \sim N(4, 0.05)$ $P[2(M_1 + \dots + M_5) > 4.20] = 0.18555 \approx 0.186 (\text{to } 3 \text{ sf})$
	<u>Alternative:</u> $M_1 + M_2 + M_3 + M_4 + M_5 \sim N(2, 0.0125)$ $P\left[M_1 + M_2 + M_3 + M_4 + M_5 > \frac{4.20}{2}\right] = P[M_1 + M_2 + M_3 + M_4 + M_5 > 2.10]$ $= 0.186 (\text{to } 3 \text{ s.f})$

Qn	Answers
9(ii)	$7D \sim N(10.5, 1.96)$ $P(7D > 10) = 0.63951 = 0.640 \text{ (to 3 sf)}$ Probability = $(0.63951)(0.18555)$ $= 0.11866 = 0.119 \text{ (to 3 sf)}$
	<u>Alternative:</u> $D \sim N(1.5, 0.2^2)$ $P\left(D > \frac{10}{7}\right) = 0.63951 = 0.640 \text{ (to 3 sf)}$
9(iii)	Let C be the price of three bags of mangoes. $C \sim N(12, 0.15)$ $7(D_1 + D_2) \sim N(21, 3.92)$ $[C + 7(D_1 + D_2)] \sim N(33, 4.07)$ $P[[C + 7(D_1 + D_2)] < 33.50] = 0.598$
	<u>Alternative notation:</u> Let J be the price of a bag of mangoes. $J_1 + J_2 + J_3 \sim N(12, 0.15)$ $7(D_1 + D_2) \sim N(21, 3.92)$ $J_1 + J_2 + J_3 + 7(D_1 + D_2) \sim N(33, 4.07)$ $P[J_1 + J_2 + J_3 + 7(D_1 + D_2) < 33.50] = 0.598$
10a)	Let X be the random variable “the run time of the lawn mower engine on a single gallon of regular gasoline”. $H_0: \mu = 300$ $H_1: \mu < 300$ Using 1 tailed test at 5% level significance level. Under H_0 , $T = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}} \sim t(14)$ $\mu = 300, \bar{x} = 290, n = 15, s^2 = \frac{n}{n-1}(\text{sample variance}) = \frac{15}{14}(20)$ $\therefore s = \sqrt{\frac{15}{14}(20)^2} = 20.70196678$ Using GC, p – value = 0.0412089348 = 0.0412 (to 3 s.f) Since p – value = 0.0412 < 0.05, reject H_0 and conclude that there sufficient evidence at 5% significance level, that the inventor has overstated the mean run time. Assume that the run time is normally distributed.

10b)	$\bar{x} = \frac{\sum(x-150)}{n} + 150 = \frac{-250}{50} + 150 = 145$ $s^2 = \frac{1}{n-1} \left[\sum(x-150)^2 - \frac{[\sum(x-150)]^2}{n} \right] = \frac{1}{49} \left[3000 - \frac{(-250)^2}{50} \right] = 35.714$ Under H_0, $Z = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}} \sim N(0, 1)$ approx. by CLT , Since H_0 is rejected in favour of H_1, Critical region: $z < -2.576$ or $z > 2.576$ $\frac{145 - \mu_0}{\sqrt{\frac{35.714}{50}}} < -2.576 \quad \frac{145 - \mu_0}{\sqrt{\frac{35.714}{50}}} > 2.576$ $\mu_0 > 147.18 \quad \mu_0 < 142.82$ $\therefore \{\mu_0 \in \square : \mu_0 < 142.82 \text{ or } \mu_0 > 147.18\}$
11(i)	$r = 0.875$ Acceptable answers: • $r = 0.875$ indicates a strong linear correlation, a linear model is appropriate. • Not appropriate as $r = 0.875$ is based on 10 data points. A linear model may not be appropriate for the population.
(ii)	 As x increases, the value of y increases by decreasing amount. Thus consistent with a model of the form $y = a + b \ln x$.
(iii)	The regression line y on $\ln x$ is $y = 3.00 + 3.19(\ln x)$ (3 s.f.) $a = 3.00$ and $b = 3.19$ (3 s.f.)
(iv)	$r \approx 0.96134 = 0.961$ (3 s.f.) This value is closer to 1 than the value 0.875 obtained earlier.
	When $y = 6$, $6 = 2.999423214 + 3.188562166(\ln x)$ $x \approx 2.562655026 = 2.56$ (3 s.f.)

Qn	Answers
12(a)	$X \sim B(10, 0.2)$, $E(X) = 20 \times 0.2 = 4$ and $\text{Var}(X) = 20 \times 0.2 \times 0.8 = 3.2$ By CLT, $\bar{X} \square N\left(4, \frac{3.2}{50}\right)$ approximately $P(3.8 < \bar{X} < 4.2) = 0.571$
12(b)	Let X be the r.v. “the number of people, out of 10, that are infected with the disease”. $X \sim B(10, 0.25)$ $P(X \geq 4) = 1 - P(X \leq 3) \approx 0.2241249 = 0.224$ (3 s.f.)
	Let Y be the r.v. “the number of people, out of 100, that are infected with the disease”. $X \sim B(100, 0.65)$ $n = 100$ is large, $np = 65 > 5$, $nq = 35 > 5$, $\therefore Y \sim N(65, 22.75)$ approximately. $P(60 \leq Y \leq 80) = P(59.5 < Y < 80.5) \approx 0.874990 = 0.875$ (3 s.f.)
	Let W be the r.v. “the number of people, out of 200, that are infected with the disease”. $W \sim B(200, 0.98)$ $n = 200$ is large, $np = 196 > 5$, but $nq = 4 < 5$, $W' \sim B(200, 0.02)$ $np = 4 < 5$, thus $W' \sim \text{Po}(4)$ approximately. $P(W \geq 196) = P(W' \leq 4) = 0.629$ (3 s.f.)